Answer Section

MULTIPLE CHOICE

1. ANS: C
2. ANS: A
3. ANS: E
4. ANS: B
5. ANS: D
6. ANS: C
7. ANS: A
8. ANS: D
9. ANS: D
10. ANS: A
11. ANS: B
    Atoms with unequal electronegativities will not share electrons equally, therefore one end of the bond ends up with a partial positive charge and the other ends up with a partial negative charge. Atoms with low electronegativities are metals, which do not bond covalently with each other.
12. ANS: B
13. ANS: C
14. ANS: C
15. ANS: C
16. ANS: D
17. ANS: B
18. ANS: B
19. ANS: C

SHORT ANSWER

20. ANS:
    Magnesium has a full s orbital. Since full orbitals are more stable than partially filled orbitals, it is more difficult to remove that last electron. Aluminum has one electron in its p orbital. Since losing this electron would leave it with a full s orbital, this electron is easily removed. Similarly, if sulfur loses an electron, it is left with a half filled set of p orbitals, which is a relatively stable configuration. As a result, it is relatively easy to remove the electron from sulfur. Since phosphorus already has the half filled orbital configuration, it is difficult (relative to sulfur) to remove an electron from it.
21. ANS:

↑↓
4s

↑↓ ↑↓ ↑↓
3p

↑↓
3s

↑↓ ↑↓ ↑↓
2p

↑↓
2s

↑↓
1s

22. ANS:

↑↓ ↑ ↑
2p

↑↓
2s

↑↓
1s

$1s^2 2s^2 2p^4$

Pauli exclusion principle states that no two electrons can have the same four quantum numbers, therefore each orbital can hold only two electrons with opposite spins. Hund's rule says that the electrons in orbitals with the same energy are half filled first before more are added. Also, the electrons in those half filled orbitals must have the same spin.

23. ANS:
Sulfur's valence shell is the third energy level. The third energy level contains s, p and d orbitals but sulfur uses only the s and p orbitals for its own electrons. As a result, it can form coordinate bonds with more than four atoms by using its empty d orbitals.

24. ANS:
- In a single bond, there is direct overlap of two orbitals.
- In a double bond, two orbitals directly overlap and two others overlap laterally.

25. ANS:

↑ __
3d

↑↓
4s

↑↓ ↑↓ ↑↓
3p

↑↓
3s

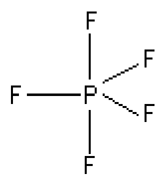
↑↓ ↑↓ ↑↓

2p

↑↓
2s

↑↓
1s

26. ANS:
volume of space in which there is a high probability of finding an electron
27. ANS:
The two bonded pairs and two unbonded pairs of electrons around the central atom and then spread out in space so that they can be as far apart as possible. Thus, they form a tetrahedral shape. The shape of water(H-O-H) ends up being bent. because the unbonded pairs of electrons are invisible.
28. ANS:



The shape is trigonal bipyramidal with 120° on the horizontal plane and 90° along the vertical plane.
29. ANS:
Linear with 180° bond angle.
30. ANS:
The delocalized electrons are free to move from one place to another, thus forming an electric current.

ESSAY

31. ANS:
Water is a small covalent compound that is highly polar. It does not conduct electricity as it is not ionic and there are no delocalized electrons. The molecule is polar because the oxygen has two lone pairs of electrons. Thus, the shape is based on a tetrahedron and the molecule is bent. Since oxygen is so electronegative, the electrons spend more time with it than the hydrogens, and the molecule ends up being polar. The hydrogens- each with partial positive charges- are then attracted to the partially negative oxygens on other water molecules. This causes cohesion between water molecules, which means that is more difficult to change it to a gas. Thus, water has an extremely high boiling point compared to other molecules like it. Also, since it is able to absorb more energy before it changes state, it has a high heat capacity and is an excellent insulator. Its polarity allows it to cling to objects easily.
32. ANS:
- Electrons repel; therefore, they will attempt to stay as far apart as possible in a molecule. To do this, bond angles must be maximized.
- Lone pairs take up more room than bonded pairs and they repel more.
- In methane, the carbon is surrounded by four hydrogen atoms. If the bonded pairs of electrons are to stay as far apart as possible in three-dimensions, they will be 109° apart. This produces a characteristic tetrahedral shape. If the molecule is like ammonia, though, with a lone pair and three atoms bonded to the central atom, the shape is still based on the tetrahedron, but, since the lone pair takes up more room, the bond angles for the N-H's will be smaller than 109°. However, the shape is named by the atoms only, therefore the shape would be trigonal pyramid. For five atoms surrounding a central atom, our choice in angles is limited by the orientation of orbitals in space. Thus, the angles are 90° and 120° instead of having all of the angles equal.